

AN INTERNSHIP REPORT ON DATA SCIENCE

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

*In partial fulfillment of the requirement for the award of degree of
Bachelor of Technology in Computer Science and Engineering*

Submitted By:

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Roll Number: 2410031349

Batch: 2024-2028

1. CANDIDATE'S DECLARATION

I, **Manthan Arora**, do hereby declare that the internship report titled “**Data Science**” has been completed by me to fulfill the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references and resources have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own original work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright or compliance violation in this regard.

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2. ACKNOWLEDGEMENT

I am using this opportunity to express my heartfelt gratitude to everyone who supported me throughout the Internship Program. I am thankful for their inspiring guidance, invaluable constructive criticism, and friendly advice during this period.

I am sincerely grateful to **Thiranex (Skill Development & Future Tech)** and its Founder & CEO, **Hariharan M**, for providing me the opportunity and structured learning platform to gain hands-on exposure in the domain of Data Science.

I express my deep sense of gratitude to my institute, **IILM University, Greater Noida**, and the faculty members of the **School of Computer Science and Engineering** for their continuous support, motivation, and encouragement.

I also express my sincere gratitude to my parents and friends for their blessings and constant support.

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4. INTERNSHIP COMPLETION CERTIFICATE

The official Data Science Internship Certificate issued by Thiranex is details below:

Organization Name	Thiranex (Skill Development & Future Tech)
Internship Domain	Data Science
Internship Duration	10 Aug 2026 to 09 Sept 2026
Certificate ID	THX-AUG1026-364
Authorized Signatory	Hariharan M (Founder & CEO, Thiranex)

5. PROJECT & INTERNSHIP DETAILS

5.1 Introduction

This report presents a comprehensive technical account of the intensive Data Science Internship undertaken at Thiranex from August 10, 2026, to September 09, 2026. The internship was completed as part of the academic curriculum for the award of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The primary objective of this internship was to bridge the gap between academic theory in statistics and algorithms, and the real-world application of machine learning, exploratory data analysis, data wrangling, and predictive modeling in enterprise pipelines.

5.2 Organization Profile

Thiranex (Skill Development & Future Tech) is an MSME-registered organization dedicated to empowering students and professionals with industry-ready skills in core emerging technologies. Thiranex designs structured training programs, hands-on projects, and virtual technical internships in domain areas including Data Science, Artificial Intelligence, Machine Learning, and Full Stack Development.

5.3 Problem Statement

While academic study provides strong theoretical background in algorithm design and discrete math, real-world data is inherently noisy, missing values, imbalanced, and unstructured. Deploying reliable data models requires an end-to-end understanding of data cleaning, feature engineering, exploratory data analysis (EDA), model tuning, and evaluation. This internship addressed these gaps by providing systematic practical exposure to the entire data science lifecycle.

5.4 Project Objectives

- To learn and implement the complete Data Science lifecycle from raw data ingestion to evaluation.
- To perform rigorous Exploratory Data Analysis (EDA) and data visualization using Python libraries.
- To gain proficiency in data preprocessing, feature engineering, and handling missing or imbalanced data.
- To construct and train supervised and unsupervised machine learning models (Regression, Classification, Clustering).
- To evaluate model performance using industry metrics (Precision, Recall, F1-Score, ROC-AUC, RMSE) and optimize performance via hyperparameter tuning.

5.5 Scope of the Project

The scope of the internship covered data collection, preprocessing pipelines, statistical modeling, exploratory analysis, feature selection, and supervised/unsupervised algorithm implementations. The tasks were executed in python lab environments using industry-standard open-source libraries.

5.6 Technologies and Tools Used

Programming Languages	Python, SQL
Data Manipulation	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning Frameworks	Scikit-Learn
Development Environments	Jupyter Notebook, Google Colab, VS Code
Version Control	Git, GitHub

5.7 Workflow & Data Pipeline Architecture

The generic data science lifecycle followed throughout the internship is structured into 5 sequential layers:

- 1. Data Ingestion & Storage (CSV, SQL Databases, JSON)
- 2. Preprocessing & Cleaning (Imputation, Normalization, Encoding)
- 3. Exploratory Data Analysis & Feature Selection (Correlation Analysis, Visual Distributions)
- 4. Model Training & Tuning (Supervised/Unsupervised Algorithms, Cross-Validation)
- 5. Model Evaluation & Reporting (Accuracy, ROC-AUC, Confusion Matrix)

5.8 Methodology & Key Modules

Phase	Module Title	Key Activities & Deliverables
Phase 1	Environment Setup & Data Manipulation	Python setup, working with NumPy arrays, Pandas DataFrames, indexing, filtering, and merging datasets.
Phase 2	Exploratory Data Analysis (EDA)	Cleaning dirty data, treating missing values/outliers, plotting distributions, heatmaps, and boxplots using Seaborn and Matplotlib.

Phase 3	Feature Engineering & Preprocessing	One-Hot / Label encoding, feature scaling (StandardScaler/MinMaxScaler), feature importance analysis, and dimensionality reduction.
Phase 4	Machine Learning Model Building	Training Linear/Logistic Regression, Decision Trees, Random Forests, and K-Means Clustering models.
Phase 5	Evaluation & Hyperparameter Optimization	Model evaluation using Confusion Matrix, ROC-AUC curves, RMSE, and hyperparameter tuning using GridSearchCV.

5.9 Key Outcomes & Learnings

- Acquired end-to-end practical skill in building data science pipelines using Python.
- Gained solid competency in data wrangling and advanced visualization to extract business insights.
- Learned to train, tune, and evaluate machine learning models for real-world predictive applications.
- Successfully completed internship requirements and earned verified certification from Thiranex.

6. BIBLIOGRAPHY & REFERENCES

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